

CONTENTS

- 26 **COMPUTING:** Where We Stand In The Race To Quantum Computing Technology
- 36 **TECH FOCUS:** Sustainable Electricity Generation Beyond Solar And Wind
- 39 **MOBILITY:** Role Of Automotive Information Technology In Autonomous Vehicles
- 44 **CAREER:** Can Non-Aerospace Engineers Work In The Aerospace Domain?
- 46 **TEST & MEASUREMENT:** How To Check Solar Systems For Efficiency And Long Life
- 49 **ARTIFICIAL INTELLIGENCE:** Can AI Rescue The World From Global Warming And Climate Change?
- 52 **HEALTHCARE:** How Hybrid Approach Can Enhance Quality Of Ultrasonic Images
- 80 **MARKET SURVEY:** Renewable Energy Has No Option But To Grow Faster
- 86 **FACTS & FIGURES:** Systems/Sub-Systems For Industry To Design, Develop And Manufacture
- 87 **FACTS & FIGURES:** Guidelines For Installation Of Innovative Standalone Solar Pumps

52ND YEAR OF PUBLICATION
electronics
 SINCE JANUARY 1969 **FOR YOU**

Next Issue

Focus Area:

Bots and Conversational Apps

- Now Energy Can Be Harvested From Shadows
- Role of Electronics in Beating Coronavirus
- Fruit Classification Using Neural Networks
- Great Opportunity in AI and ML Hardware Sector

EDITOR : Ramesh Chopra
EDITORIAL : Editorial Secretary
CORRESPONDENCE Phone: 011-26810601; E-mail: editsec@efy.in
 (Technical queries: efylab@efy.in)

SUBSCRIPTIONS & MISSING ISSUES : Phone: 011-40596600
 E-mail: support@efy.in

BACK ISSUES, BOOKS, KITS, COMPONENTS, etc : Kits'n'Spares, New Delhi
 (Prop: EFY Exports Pvt Ltd)
 Phone: 011-26371661
 E-mail: info@kitsnspares.com

NEWSSTAND DISTRIBUTION : Ph: 011-40596600, 8800295612
 E-mail: efycirc@efy.in

ADVERTISEMENTS

NEW DELHI (HEAD OFFICE) : Ph: 011-40596600
 E-mail: efyenq@efy.in

MUMBAI : Ph: 8291852259, 9867689037
 E-mail: rmwest@efy.in

BENGALURU : Ph: 080-25260023/25260394, 8800295604
 E-mail: rmosouth@efy.in

JAPAN : Tandem Inc., Ph: 81-3-3541-4166
 E-mail: japan@efy.in

SINGAPORE : Publicitas Singapore Pte Ltd
 Ph: +65-6836 2272; E-mail: singapore@efy.in

TAIWAN : J.K. Media, Ph: 886-2-87726780 ext. 10
 E-mail: taiwan@efy.in

UNITED STATES : E & Tech Media
 Ph: +1 860 536 6677; E-mail: usa@efy.in

EUROPE : ASA Media, UK
 Ph: +44 1727 765542; C: +44 7910 073565
 E-mail: alastair@asa-media.com
 pms plantenberg media service GmbH, Germany
 Ph: +49 89 5507 9909
 Mobile: +49 170 4828426; E-mail: germany@efy.in

Printed, published and owned by Ramesh Chopra. Printed at International Print-o-Pack Ltd, C-4 to C-11, Hosiery Complex, Phase-II Extension, NOIDA-201305, Gautam Budh Nagar, Uttar Pradesh, on the first day of each month and published from D-87/1, Okhla Industrial Area, Phase-1, New Delhi 110020. Copyright 2020. All rights reserved throughout the world. Reproduction of any material from this magazine in any manner without the written permission of the publisher is prohibited. Although every effort is made to ensure accuracy, no responsibility whatsoever is taken for any loss due to publishing errors. EFY will not be responsible for any wrong claims made by an advertiser. Disputes, if any, will be settled in a New Delhi court only.

Courier delivery within India @ Rs 50 per copy

e-magazine subscriptions within India are available at half the rates mentioned here.

Overseas rates for each e-magazine in US\$:
 1 year: \$12; 3 years: \$33; 5 years: \$50 only

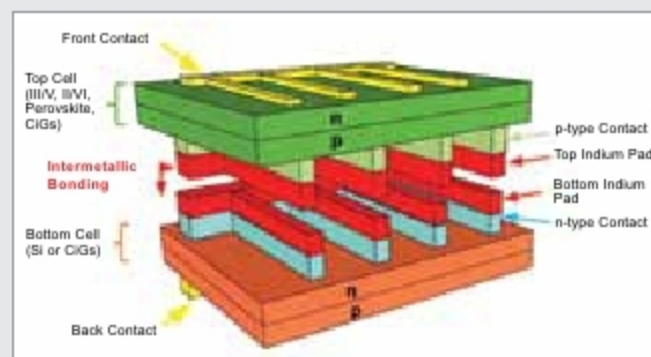
Please send payments only in favour of **EFY Enterprises Pvt Ltd** through demand draft/cheque/money order to D-87/1, Okhla Industrial Area, Phase 1, New Delhi 110020, or online at: <http://subscribe.efyindia.com> or scan the QR code.


www.EFYMag.com

RENEWABLE ENERGY

Six-Junction Solar Cells Can Reach Astounding Efficiency Of 47 Per Cent

30



INTERVIEW

- 56 **“Data Quality Is A Top Barrier To Successful Adoption Of AI”**
 Johanna Pingel, marketing manager, deep learning, MathWorks
- 57 **“We Do Not Mentor Startups, We Mentor Individuals Worthy Of Creating Startups”**
 Esha Tiwary, general manager, Entrepreneur First (EF), India

DO-IT-YOURSELF

- 58 Social Distancing Alarm Cap
- 60 Solar Day Lamp With Battery Backup
- 63 Hand Gesture Based Wireless AC Light Dimmer
- 65 Automatic Water Dispenser
- 67 Multilingual Home Automation System Using Google Assistant And Raspberry Pi
- 71 MATLAB GUI Based Ultrasonic Liquid Level Monitoring System

SMART WORLD

- 18 Smart Cybersecurity For Emerging Technologies
- 20 Organic Semiconductors And LEDs
- 22 The Importance Of Smart Speakers In Modern Era
- 24 Intellectual Property Rights In India

STARTUP ZONE

- 76 AI For Solar Asset Management
- 77 XR Services Transforming Businesses
- 78 IoT-Enabled Device For Secure Deliveries
- 78 Enhanced Techniques For Reviving Geothermal Sector

eSTYLE

- 90 **FIRST LOOK:** Some New Gadgets
- 92 **GAMES:** The Growing Popularity Of eSports Amidst Covid-19 Pandemic
- 94 **BUYERS' GUIDE:** Roof-Top Solar Panels For Homes And Factories

Regulars

- 08 **Feedback:** Your Suggestions
- 10 **Q&A:** Things You Wanted To Know!
- 12 **Tech News:** Technology Updates
- 73 **Industry News:** Industry Updates
- 83 **New Products**
- 88 **Software and Apps**